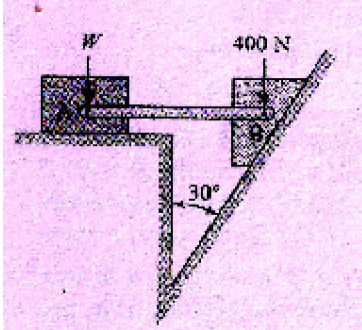
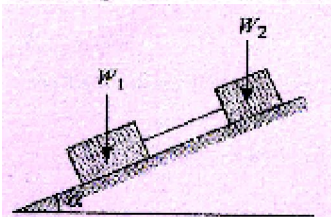


Model questions on Friction for practice(MOD-3)

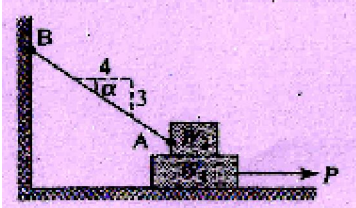
Q1. Two blocks connected with a tie rod are shown in Fig. If $\phi_B = 15^\circ$ and $\mu_A = 0.40$, find the smallest value of W for the equilibrium of the system.



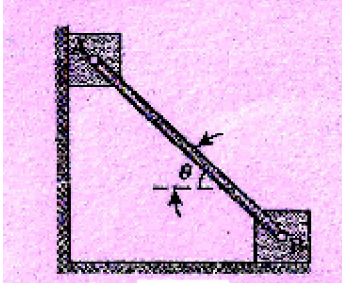
Q2. Two blocks of weights W_1 and W_2 connected with a string rest on a rough incline, as shown in Fig. If the coefficients of friction are 0.2 and 0.3 for the blocks, respectively, and $W_1 = W_2 = 50$ N, find the value of α for which sliding will impend.



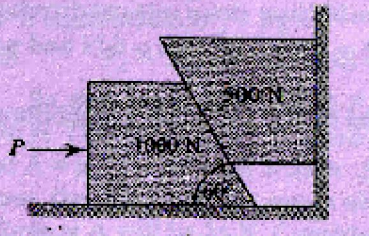
Q3. A block of weight $W_1 = 200$ N rests on a horizontal surface and supports on top of it another block of weight $W_2 = 50$ N. The block W_2 attached with vertical wall by a string AB, is shown in Fig. Find the amount of horizontal force P , applied to the lower block necessary for impending slipping. The coefficient of friction for all contiguous surfaces is 0.3.



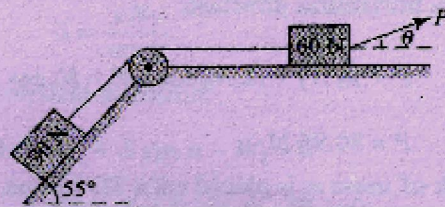
Q4. Two identical blocks, A along the vertical wall and B on the floor are kept stationary with the help of a connected strut, as shown in Fig. If sliding impends when $\theta = 45^\circ$, determine the coefficient of friction, assuming it to be same at both the floor and the wall.



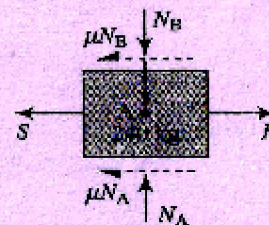
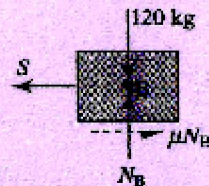
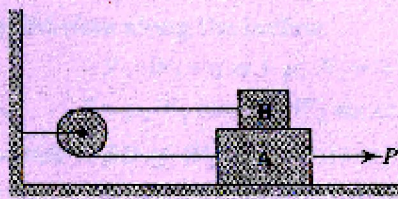
Q5. Referring to Fig. the coefficients of friction are as follows: 0.25 at the floor, 0.30 at the wall, and 0.20 between interfaces of the blocks. Find the minimum value of a horizontal force P applied to the lower block that will hold the system in equilibrium.



Q6. Referring to Fig. determine the least value of P to cause motion to impend rightwards. Assume the pulley to be frictionless and coefficient of friction of all contiguous surfaces is 0.20.



Q7. Block A weighing 240 kg resting on a rough floor supports block B weighing 120 kg as shown in Fig. Both the blocks are connected with a rope passing over a smooth pulley. Compute the magnitude of force P at impending motion and the tension induced in the rope, if the coefficient of friction for all contiguous surfaces is 0.3.

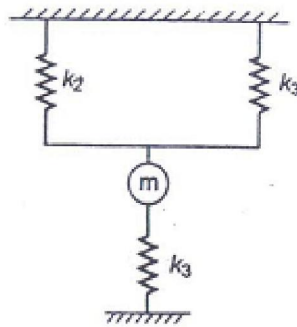


Model Questions on Natural Frequency of a Spring Mass System

- Q8.** A vibration system consists of a mass m and a spring of stiffness k . It has natural frequency of 12 Hz. The natural frequency decreases by 2 Hz. When an extra mass of 2 kg is attached to m . Find k and m .

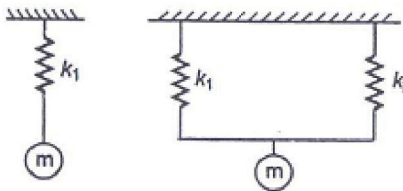
(Ans 25.783 N/mm)

- Q9.** Find the natural frequency of the system shown in Figure 1 if $k_1 = k_2 = 1500$ N/m, $k_3 = 2000$ N/m and $m = 5$ kg.



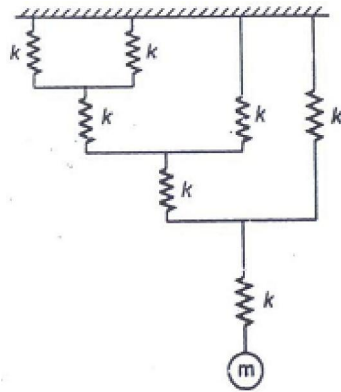
(Ans 5.03 Hz)

- Q10.** A Spring mass system (k_1 , m_1) has a natural frequency of f_1 as shown in figure 2. Another spring k_2 is connected in parallel to k_1 and the natural frequency increases by 30%. Find k_2 .



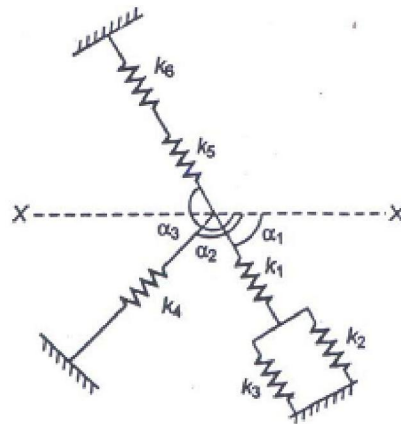
(Ans $k_2 = 0.69k_1$)

- Q11.** Calculate the natural frequency of the system shown if $k = 2 \times 10^5$ N/m and $m = 20$ kg.



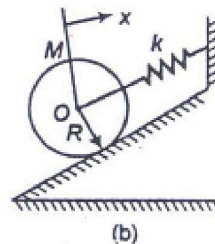
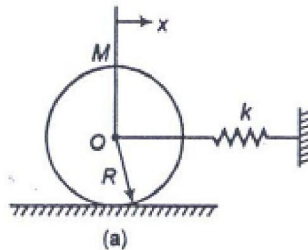
(Ans $f_n = 12.5 \text{ Hz}$)

- Q12.** Find the equivalent spring constant and natural frequency for oscillation of mass m in the direction $x-x$ as shown in figure.



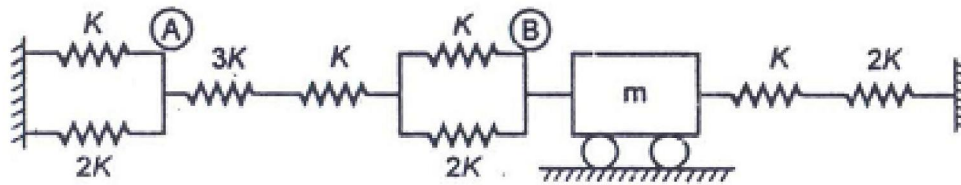
(Ans $f_n = \frac{1}{2\pi} \sqrt{\frac{k_e}{m}}$ $k_e = \frac{k_1 (k_2 + k_3)}{k_1 + k_2 + k_3} \cos^2 \alpha_1 + k_4 \cos^2 \alpha_2 + \frac{k_5 k_6}{k_5 + k_6} \cos^2 \alpha_3$)

- Q13.** A circular cylinder of mass ' m ' and radius ' R ' is connected by a spring of stiffness ' k ' as shown in Fig. (a). If it is free to roll on the rough surface which is horizontal without slipping. Find its natural frequency. What will happen to frequency if the system is placed on an inclined as shown in Fig. (b).



Ans. $f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3M}} \text{ CPS.}$

Q14. Determine the equivalent stiffness of the system as shown in Fig.



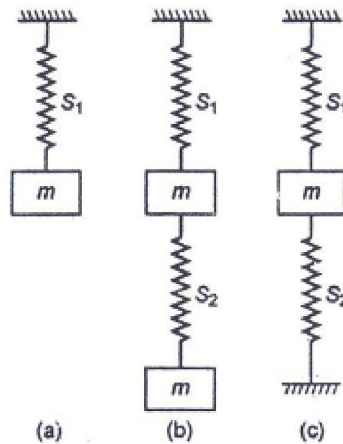
$$k_{eq} = 1.18 \text{ N/m}$$

A

Q15. Determine the equivalent spring stiffness and the natural frequency of the following vibrating systems when

- (a) mass is suspended to a spring
- (b) mass is suspended at the bottom of two springs in series
- (c) mass is fixed in between two springs

Take $S_1 = 5 \text{ N/mm}$, $S_2 = 8 \text{ N/mm}$, $m = 10 \text{ kg}$,



Ans a) $f_n = 3.56 \text{ Hz}$

b) $f_n = 2.79 \text{ Hz}$

c) $f_n = 5.74 \text{ Hz}$